

DR. PANDA 2.0 AI MEDICAL INTELLIGENCE PLATFORM

Interactive Panda Doctor Screening Persona, Medical Report Parsing, and Care Triage System

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Institution: MIT Manipal

Domain: AI Healthcare & Medical NLP

Stack: FastAPI, Streamlit, Scikit-learn, PyTorch

KEY PLATFORM CAPABILITIES

Interactive

Dr. Panda Persona

Sub 200 ms

Symptom Query Response

Parser

Lab Report Reader

3 Tiers

Triage Priority Rating

SYSTEM ARCHITECTURE & FEATURES

Dr. Panda 2.0 is an expanded medical screening system engineered to provide supportive symptom analysis, parse uploaded medical reports for health biomarkers, and compute care triage recommendations across Emergency, Moderate, and Low priority tiers.

- Conversational Panda Doctor UI: Interactive Streamlit interface featuring a supportive doctor character persona
- Medical Report Reader: Extracts lab test parameters (glucose, cholesterol, hemoglobin, WBC) and determines hospital visit urgency
- Triage Engine: Recommends specific patient actions including home care, rest, physician scheduling, or emergency room visits
- AI Exclamation Disclaimer: Prominently featured across UI and API endpoints emphasizing suggestive educational bounds

PUBLIC DATASETS INCLUDED

Dataset File	Record Description	Primary Medical Purpose
data/medical_qa_dataset.csv	Public Medical Screening QA Pairs	Symptom similarity matching and clinical screening
data/lab_test_benchmarks.csv	Lab Test Biomarker Reference Values	Medical report text parsing and threshold evaluation
data/symptom_triage_rules.csv	Triage Urgency Rules	Rating hospital visit vs home care requirements